

What Drives University Rankings? A Data-Driven Look

A self-directed analysis of Executive Education-style ranking data, built using SQL, Python and Power BI to mirror the data mining and reporting work described in the Aalto EE Data Analyst/Engineer role.

Methodology

A public world university rankings dataset (2012–2015, 100+ institutions per year) was structured into a relational database using **SQL** (SQLite), queried to surface ranking patterns, analysed statistically in **Python** (pandas) to test which ranking criteria most influence outcomes and visualised as an interactive two-panel dashboard in **Power BI**.

Key Findings

1. Faculty quality is the strongest predictor of overall score

Across seven ranking criteria tested (quality of faculty, quality of education, alumni employment, publications, influence, citations, patents), **quality of faculty showed the strongest relationship with final score** (correlation coefficient 0.71), ahead of research output measures like publications and citations (both ~0.52) and alumni employment outcomes (0.48). This suggests ranking bodies may weight faculty reputation more heavily than research volume, a useful benchmarking signal for Aalto EE when interpreting its own FT ranking position.

2. Quality outperforms scale: fewer, stronger institutions can beat larger systems

Singapore and Israel led all countries on average institutional score in 2015 despite having only 2 and 7 ranked universities respectively ahead of the USA (229 institutions) and the UK (65). This indicates average ranking score rewards selectivity and consistency, not the size of a national higher-education system.

3. The Netherlands, Denmark and Norway show the strongest upward momentum (2012–2015)

Tracking average score by country over time shows Nordic and Dutch institutions improving fastest, while traditionally strong performers such as Switzerland and Italy show a relative decline over the same period, a trend worth monitoring in ongoing peer benchmarking.

Tools & Deliverables

Tool	What it demonstrates
SQL (SQLite)	Relational data modelling, filtering, grouping, aggregate queries
Python (pandas)	Statistical correlation analysis, multi-year trend calculation
Power BI	Interactive two-visual dashboard (score by country, universities ranked per country)

Full SQL queries, Python scripts, dataset and dashboard file available on request / GitHub link below.

github.com/Temitope19193/aalto-ranking-analysis.

Prepared for Aalto University Executive Education – Data Analyst/Engineer application